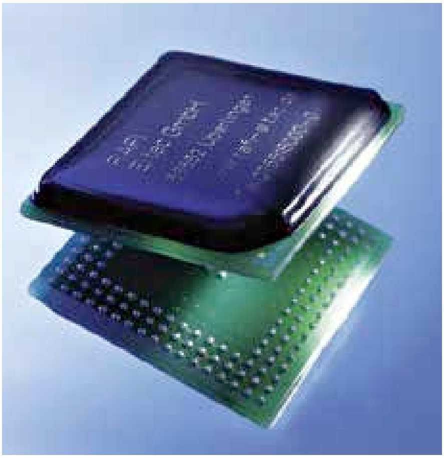
A method of fabrication



Packing in all that power.

Multiple technologies can be used for the manufacturing of SoCs. To detail all of them would be beyond the scope of this article. But since the mode of fabrication gives insight into what goes behind the scene in making a chip, it's worth knowing in detail at least about one key mode of fabrication.

Standard cell

Standard cell methodology is used to design application-specific integrated circuits or ASICs. This incorporates mostly digital-logic features. The method is also a great example of design abstraction- which means a low-level very large scale integration or VLSI layout gets encapsulated in an abstract logic representation (for example, the NAND gate).

Standard cells belong to the general methodology class called cell-based methodology. It demarcates the designers' functions. For instance, one designer could focus on the logical function facet of the design while another would focus on the physical or implementation facet.

Thanks to the standard cell method, designers have been able to scale up ASICs from rather simple and single function ICs that involve many thousand gates to highly intricate SoC devices that include many millions of gates.

How it's constructed?

A standard cell essentially means a bunch of transistors and interconnected structures which provide a Boolean logic function. Examples include OR, AND, XOR and XNOR inverters. Or it could provide a storage function- like in the case of a latch or flipflop. The simplest of these cells represent the elemental NOR, NAND and XOR Boolean function. But that's not to say that cells of a greater complexity aren't commonly used- for example, the D-input flipflop or the 2-bit full adder.